CLOUD SECURITY — PRACTICAL-BASED LEARNING

PBL 4: Create a VPC with Public and Private Subnets

Standalone practical report

1. Aim

Create an AWS VPC with public and private subnets, deploy web servers in each, and test whether the public server can retrieve the private website without direct internet exposure of the private server.

2. Problem Statement

Create a VPC, public and private subnets, attach an Internet Gateway to the public route, deploy web servers, and configure security groups so the public server can retrieve the private page over HTTP while direct public access to the private server remains blocked.

3. Tools and Technologies

- AWS VPC and EC2
- Internet Gateway and route tables
- Security Groups
- Amazon Linux 2023
- Apache httpd, curl and Python HTTP server as a proposed fallback
- EC2 Instance Connect Endpoint

4. Actual VPC and Subnet Configuration

Component	Recorded configuration
VPC	Cloud-Security-PBL4 / pbl4
VPC CIDR	10.0.0.0/16
Public subnet	PBL4-Public-Subnet — 10.0.1.0/24
Private subnet	PBL4-Private-Subnet — 10.0.2.0/24
Internet Gateway	PBL4-IGW
Public route table	PBL4-Public-RT; 0.0.0.0/0 → IGW
Private route table	PBL4-Private-RT; local VPC route only recorded
Private instance	PBL4-Private-Web
Private IP	10.0.2.68
Operating system	Amazon Linux 2023
Private access	EC2 Instance Connect Endpoint

The private instance was recorded without a public IP. Verify current AWS console values before reusing any configuration.

5. Procedure

Step 1 — Create VPC and subnets

Create the VPC with 10.0.0.0/16, public subnet 10.0.1.0/24 and private subnet 10.0.2.0/24.

Step 2 — Configure routes

Attach PBL4-IGW to the VPC. Configure PBL4-Public-RT with 0.0.0.0/0 to the IGW. PBL4-Private-RT was recorded with the local VPC route only.

Step 3 — Test public server

The public server's local curl returned: **<h1>PBL4 Public Web Server</h1>**. This proves the recorded local HTTP response, not external internet accessibility.

Step 4 — Launch private server

Launch PBL4-Private-Web in the private subnet. Its recorded private IP is 10.0.2.68 and no public IP was recorded.

Step 5 — Create EC2 Instance Connect Endpoint

Create and use the endpoint for private administrative access. This connection path does not automatically provide general outbound internet access to the private instance.

Step 6 — Install Apache

```
sudo dnf install -y httpd
```

The installation timed out while reaching the package repository.

Step 7 — Create the webpage

```
sudo mkdir -p /var/www/html\necho '<h1>PBL4 Private Website</h1><p>Accessed through the public server.</p>'
```

These commands were identified as the correction for the missing directory and malformed command invocation. Final successful file creation was not confirmed by the recorded evidence.

Step 8 — Proposed Python server workaround

```
sudo nohup python3 -m http.server 80 --directory /var/www/html --bind 0.0.0.0 >/tmp/pbl4-http.log 2>&1 &\ncu
```

This was proposed as a temporary alternative if Python was already installed. Successful execution is not confirmed.

Step 9 — Test from public server

```
curl --connect-timeout 5 http://10.0.2.68
```

The recorded request timed out or failed to connect.

6. Errors Encountered and Solutions

dnf package installation timeout

The private instance timed out while contacting its package repository. The lack of a recorded internet default route in the private subnet is a likely cause. Options include a NAT Gateway with an appropriate private route, an approved reachable package source, or a preconfigured image. A NAT Gateway may add cost. The Apache installation was not confirmed successful.

tee: invalid option -- 'y'

Commands were pasted or combined in a way that passed an unintended option to tee. Run commands separately and do not append dnf options to tee.

No such file or directory for /var/www/html/index.html

The target directory did not exist. Create it first using sudo mkdir -p /var/www/html, then write index.html. Successful file creation remains unverified without terminal output.

curl connection failure from public server

The request to http://10.0.2.68 failed or timed out. Check that the private instance is running, the HTML file exists, the HTTP server is listening on port 80, and the private instance's security group allows HTTP from the security group actually attached to the public server.

EC2 Instance Connect Endpoint versus internet access

The endpoint enabled private administrative access, but does not provide general package-repository connectivity. For SSH, the private instance security group should allow TCP 22 from the endpoint security group; the endpoint must be able to reach the instance on that port.

Conflicting public security-group IDs

The history contains sg-083ed0727c917a4b2 (launch-wizard-4) and sg-06811dc120997deff (pbl4 public sg). Do not assume they are the same or that both are attached to the public instance. Check the public instance's actual attached security group and reference that group as the source for private inbound HTTP TCP 80.

7. Result

The VPC/subnet design and route-table intent are documented. The public server's local HTTP test returned the expected PBL4 Public Web Server heading. The EC2 Instance Connect Endpoint was created and used to connect to the private instance.

The private Apache installation timed out; webpage creation encountered tee and missing-directory errors; and the final curl request from the public server failed or timed out.

FINAL STATUS: PARTIALLY COMPLETED — the final private website retrieval is UNVERIFIED. The available history does not establish a successful curl response containing the private page or a successful direct-public-access denial test.

8. Security Measures

- Separate public and private subnets.
- Keep the private instance without a public IP.
- Use the EC2 Instance Connect Endpoint for controlled private administration.
- Allow private-instance SSH only from the endpoint security group.
- Allow private-instance HTTP only from the actual public server security group.
- Keep the IGW default route in the public route table; add private outbound routing only when needed.
- Monitor charges for EC2, endpoints and any NAT Gateway; delete unneeded resources after the lab.

9. Conclusion

This practical documents a VPC with public and private subnets, an Internet Gateway, separate route tables and private administrative access through an EC2 Instance Connect Endpoint. Troubleshooting covered a dnf timeout, tee command errors, a missing web-root directory, a failed curl request and security-group source verification.

The final requirement—to retrieve the private website from the public server using curl while keeping the private server inaccessible directly from the internet—remains unverified. Add successful curl output and evidence of the private server's lack of public exposure before claiming full completion.

10. Screenshots Checklist

- ■ VPC details and CIDR 10.0.0.0/16.
- ■ Public and private subnet CIDRs.
- ■ Internet Gateway attachment.
- ■ Public route table with 0.0.0.0/0 to IGW.
- ■ Private route table showing local route only.
- ■ Public server local HTTP response.
- ■ Private instance details, private IP and no public IP.
- ■ EC2 Instance Connect Endpoint details.
- ■ dnf timeout output.
- ■ tee invalid-option and missing-directory errors.
- ■ Successful file creation and private localhost curl, if available.
- ■ Private SG HTTP rule sourced from the actual public server SG and SSH rule from endpoint SG.
- ■ Final public-to-private curl output; if it fails, label result unverified.
- ■ Evidence of direct public access being blocked.
- ■ Resource cleanup evidence.